Topic architecture

Topic	Contents	Key	Type
`patient-profile`	patient_id, age, maladie, médecin_responsable	`patient_id`	Compacted (reference data)
`wearable-vitals`	heart_rate, ecg, spo2, body_temperature, fall_detection	`patient_id`	High-frequency stream
`smarthome-context`	alone, presence_room, dose_taken, room_temperature, smart_speaker_dcb	`patient_id`	Contextual stream
`device-connectivity`	per-device connection state (validates whether events are real or should be discarded)	`device_id`	Status stream

Message schemas (JSON)

`patient-profile` — key = `patient_id`, produced once per patient / on update:

```json
{
  "patient_id": "P001",
  "age": 78,
  "maladie": "insuffisance cardiaque",
  "medecin_responsable": "Dr. Ben Salah",
  "updated_at": "2026-07-14T10:00:00Z"
}
```

`wearable-vitals` — key = `patient_id`, one message per reading:

```json
{
  "patient_id": "P001",
  "device_id": "watch-001",
  "timestamp": "2026-07-14T10:15:32Z",
  "heart_rate": 128,
  "ecg": [0.02, 0.05, 0.9, -0.1, 0.03],
  "spo2": 91,
  "body_temperature": 37.4,
  "fall_detection": 1
}
```

`ecg` as a short numeric array (or waveform ID pointing to a stored trace) is more practical than embedding a full diagram — decide based on what your Tier 1/Tier 2 code actually needs to consume.

`smarthome-context` — key = `patient_id`:

json

```json
{
  "patient_id": "P001",
  "timestamp": "2026-07-14T10:15:30Z",
  "alone": 1,
  "presence_room": "living_room",
  "dose_taken": 0,
  "room_temperature": 29,
  "smart_speaker_max_dcb": 72
}
```


Use `"presence_room": "none"` when nobody's detected anywhere, rather than a separate boolean — keeps presence as one field.

`device-connectivity` — key = `device_id`:

json

```json
{
  "patient_id": "P001",
  "device_id": "watch-001",
  "device_type": "wearable",
  "timestamp": "2026-07-14T10:15:32Z",
  "connected": 1
}
```